Jack Agterdenbos

MSc Aerospace Engineering · TU Delft · AI, Machine Learning & Mathematical Optimisation

jackagterdenbos@gmail.com · linkedin.com/in/jack-agterdenbos-2a74b6241 · jackbehindwood.github.io/portfolio · github.com/JackBehindWood · Bussum, Netherlands · Dual Dutch-British Citizen

PROFILE

MSc Aerospace Engineering (Sustainable Air Transport) from TU Delft, completed October 2026. I build working models and software end to end, from a forecasting pipeline at KLM Cargo to a hardware prototype at Juvoly. Available now, preferred start January 2027, earlier by agreement.

EDUCATION

MSc Aerospace Engineering, Sustainable Air Transport · TU Delft · *Sep 2024 – Oct 2026*

Thesis: Trajectory-Based Aircraft Landing Problem (TBALP) · TU Delft repository

- Formalised the TBALP under a Goal-Conditioned Hierarchical Reinforcement Learning framework: a Constrained Position Shifting planner assigns runways and arrival times, an Extra Trees surrogate gives fast arrival-time estimates and a Soft Actor-Critic agent adapts the flight path to meet them.
- Python, PyTorch, Stable-Baselines3 and bluesky-gym; trained and evaluated across all twelve runway ends of Amsterdam Schiphol.

BSc Aerospace Engineering · TU Delft · *2021 – 2024* · **Semester Exchange, Business Administration · UC3M Madrid** · *2023 – 2024* · **VWO · Vituscollege** · *2021*

EXPERIENCE

Intern, Cargo Control Center · KLM Cargo · *Sep 2025 – Jan 2026*

- Built an end-to-end ULD (Unit Load Device) availability forecasting model in Python, from data ingestion and pipeline design to a Prophet and scikit-learn model tuned with Bayesian optimisation, as the baseline for later allocation work.

Student Prototype Developer · Juvoly · *Jul 2025 – Present*

- Sole developer of a portable wireless audio recorder for clinical consultations: C++ firmware on an ESP32 with AES-256 encryption and secure local storage, integrating with Juvoly's AI transcription pipeline. Also built a Python and Google Firebase website for transcribing the audio.
- Took it from concept to working prototype independently, intended for Juvoly's platform spanning 50+ medical specialisms in GP, hospital and occupational health settings.

Private Tutor, Maths, Physics & Chemistry · DeBijlesStudent B.V. · *Nov 2021 – Present*

- One-to-one tutoring at secondary and pre-university level.

Earlier roles: Front Office Employee, Speelpark Oud Valkeveen (Jul–Aug 2022) · Grocery Delivery Driver, Picnic Technologies (Jun–Aug 2021) · Stock Clerk, Albert Heijn (May 2017–Dec 2020)

SKILLS

Programming: Python, C++, SQL, Git

Machine learning: Reinforcement learning (SAC, goal-conditioned), supervised and unsupervised learning, surrogate modelling, scikit-learn, PyTorch, XGBoost

Optimisation & operations research: MILP and LP modelling, Gurobi, column generation (university group projects)

Statistics & forecasting: Statistical testing (Diebold-Mariano, Wilcoxon), time-series forecasting (Prophet), Bayesian optimisation, Monte Carlo simulation

Data, tools & embedded: pandas, NumPy, SciPy, Plotly, Streamlit, FastAPI, Excel, PowerPoint, ESP32 firmware, Firebase

Languages: Dutch native · English native / bilingual · German limited working proficiency · Spanish elementary